



Faculty of Design and Creative Technology

School of Engineering, Computer and Mathematical Sciences

Department of Electrical and Electronic Engineering

ENGE707 Data Engineering and AI

Project Phase-2 Report 30% | Wizard of ODDs

Australia & New Zealand Road Crash Dataset

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(BELOW IS EXTRA INFO)

ENGE707_ProjectReportPhase2 | WizardOfOdds

https://github.com/IrtiyaHana/ENGE707_WizardsofOdds - Link to Repository

<https://www.kaggle.com/datasets/mgray39/australia-new-zealand-road-crash-dataset> - Link to Kaggle

worth 30%

Each group must submit:

- Final report: 5,000--6,500 words, excluding references and appendices.
Source code or Jupyter Notebook.
GitHub repository link and commit-history evidence.
Contribution statement, including a brief individual reflection from each group member.
Cleaned and feature-engineered dataset, or clear dataset reproduction instructions.
Turnitin similarity report with a similarity score not exceeding 20%.

must make 2 commits per week.

demo (mark reduction of up to 70%) – like feedback session

ai assistance for code only, not for report.

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Task 1: Project Phase-1 Summary

Students must briefly summarize the Phase-1 project. Students are required to:

- Restate the selected problem, dataset, target variable, and prediction objective.
- Summarize the dataset source, number of records, number of features, and main data characteristics.
- Briefly describe the main data-quality issues identified in Phase-1.
- Explain why the problem is suitable for an LCS-based solution.

Road traffic accidents across Australia and New Zealand present a severe public health and economic challenge, despite many safety and awareness initiatives. Therefore, transport safety agencies must shift from reactive emergency responses to proactive, data-driven interventions. The primary objective of this project is to build a predicative classification pipeline that identifies whether a fatal road crash involves a Youth driver or occupant aged 17 – 25 (Target_Age_Group = 1) versus an Older demographic aged 26 and above (Target_Age_group = 0). This model excludes individuals under 17 to ensure the target variable reflects active driver risks profile rather than passive passengers. Instead of predicting the actual occurrence of a crash, the system analyzes situational, temporal, and environmental factors including time of day, day of week, speed limits, and vehicle counts to identify high-risk demographic patterns.

The analysis relies on the Australia & New Zealand Road Crash Dataset (ardd_fatalities.csv), which comprises of 55,360 total records across 24 feature columns. To build a reliable, predictive baseline, we selected key variables including State, Speed Limit, Crash Type, vehicle involvement (Bus, Heavy Rigid Truck, and Articulated Truck), Day of the week, and Time of day. However, initial data analysis in Phase 1 revealed several data quality issues that could negatively impact model training.

Specifically, a trailing CSV comma delimiter introduced an artifact column (Unnamed: 23) containing 100% missing values, while Australian regional columns (SA4 Name 2021, National LGA Name 2021, National Remoteness Areas) exhibited roughly 80% missingness because New Zealand crash events do not map to Australian Bureau of Statistics boundary codes. Furthermore, missing data across key attributes including speed limits, age, and truck involvement flag was routinely encoded using the sentinel value -9. The raw dataset also contains 168 duplicate rows and inconsistent text casing. Additionally, older drivers experienced three times as many crashes as younger demographics, creating a severe class imbalance. This significant difference may cause the model to be bias towards the majority group.

To address these challenges, Learning Classifier Systems (specifically eLCS) were selected as the core algorithmic framework. Unlike neural networks that offer limited transparency, eLCS combines evolutionary algorithms and reinforcement learning to generate readable IF-THEN decision rules. As shown in Phase 1, predicting youth crashes depends on subtle feature interactions (e.g., weekend midnight hours combined with specific speed zones). eLCS can capture these overlapping non-linear rules while providing transport authorities with transparent, actionable conditions.

Task 2: Original LCS System on Raw Dataset

Students must run an original LCS system on the raw or original dataset before making improvements. Students are required to:

- Select an appropriate LCS variant, preferably eLCS.
- Use the original LCS code without algorithmic modification.
- Run the original LCS system on the raw or minimally processed dataset.
- Report the selected LCS parameters.
- Record the baseline performance results.
- Explain any minimal processing required to make the dataset compatible with the LCS code.

Baseline eLCS Configuration

The educational Learning Classifier System (eLCS) from the open source skeLCS library was used as the main rule-based learning framework, with no custom changes to its algorithms. The baseline experiment ran with the standard settings: a maximum rule population of 500, 5,000 learning iterations, and a specification probability (p_{spec}) of 0.5. Using the unmodified eLCS on the raw dataset ensures that any later improvements in performance result solely from data cleaning, feature engineering, or class balancing, rather than from altering the LCS rule-generation process.

Minimal Processing and Dataset Preparation

To prepare the Australian/New Zealand Road Deaths Database (`ardd_fatalities.csv`) for use in the skeLCS engine, only minimal preprocessing was applied to convert the dataset into a compatible numerical feature matrix. Records with invalid age codes (e.g., -9) or individuals younger than 17 were removed to focus exclusively on licensed drivers and riders. A binary target variable, `Target_Youth`, was then created, assigning 1 to young drivers aged 17 – 25 and 0 to mature drivers aged 26 and above. Non-essential columns such as *Unnamed: 23* were dropped, and categorical fields including state, crash type, and day-of-week were encoded into simple numeric categories. No additional preprocessing steps such as missing-value imputation, feature scaling, outlier removal, or class balancing were performed at this stage.

Baseline Model Performance Evaluation

Evaluating the baseline eLCS model using an 80/20 stratified split produced an overall accuracy of 72.07%, a weighted precision of 51.94%, a weighted recall of 72.07%, and a weighted F1-score of 60.37%. Despite these nominal metrics, the confusion matrix – 7,358 true negatives, 0 false positives, 2,852 false negatives, and 0 true positives shows that the model completely failed to identify the minority class. Because the raw dataset contains a severe 3:1 class imbalance and substantial missing data, the baseline eLCS model defaulted to predicting only the majority class (older drivers) for every test instance. This behavior inflates accuracy to the natural majority class proportion while yielding 0% recall for youth-related crashes. The result clearly demonstrates the need for the data cleaning, imputation, and resampling procedures introduced in Task 3.

Task 3: Data Processing and Feature Engineering

Students must improve the quality of the dataset before developing the improved system and make it suitable for working with LCS. Students are required to:

- Address missing values, duplicate records, invalid values, inconsistent categories, and incorrect data types.
- Investigate and handle outliers where appropriate.
- Encode categorical variables and scale, discretize, or transform numerical variables where required.
- Address class imbalance where relevant.
- Apply feature engineering, feature selection, or dimensionally reduction where justified.
- Clearly explain how preprocessing is expected to improve LCS performance.
- Ensure that preprocessing avoids data leakage.

Data Cleaning and Train-Test Isolation Protocols

Data cleaning steps were applied to prevent corruption before model training. Placeholder values for missing data (-9, -9.0, "Unknown") were converted to NaN, and non-target records such as passengers younger than 17 were removed. To avoid data leakage, the dataset was split into an 80/20 train-test partition before any preprocessing. Imputation rules were learned only from the training set: categorical missing values were filled using the training-set mode, and numerical missing values were filled using the training-set median. These trained imputers were then applied to the test set to ensure the model never accessed information from the test data.

Feature Engineering and Encoding

Because Learning Classifier Systems require discrete condition matching, all categorical variables were ordinal encoded using mappings learned only from the training data. Domain-specific features were engineered to improve rule discovery, including an `Is_Night` indicator derived from crash times to highlight late-night risk patterns. Numerical variables were cleaned and constrained to valid ranges to prevent invalid values during eLCS evolutionary operations. High-cardinality noise features and structural artifacts were removed to reduce the classifier's search complexity.

Resolving Class Imbalance & Leakage Prevention

To address the severe 3:1 majority-class bias documented in Task 2, Synthetic Minority Over-Sampling Technique (SMOTE) was applied exclusively to the training data ($X_{\text{train}}, y_{\text{train}}$), expanding the minority Youth class to achieve an exact 1:1 balance before rule discovery. The test dataset ($X_{\text{test}}, y_{\text{test}}$) remained in its original imbalanced form to ensure that final performance metrics reflect unbiased, real-world generalization.

Theoretical Justification for eLCS Enhancements

These preprocessing steps directly addressed the core limitations of the baseline eLCS run. By eliminating sentinel noise, eLCS is prevented from generating meaningless condition predicates based on missing value codes. Ordinal encoding ensures that genetic mutation and crossover operate over valid numerical intervals. Most importantly, rebalancing the training distribution forces the reinforcement learning reward structure to treat Youth crash instances with equal importance, preventing majority-class collapse and enabling the discovery of precise rules for young-driver risk patterns.

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Task 4: Improved LCS-Based System

Students must improve the LCS code, LCS configuration, or machine learning pipeline to develop a novel improved system. Possible improvements include:

- Hyperparameter tuning.
- improved preprocessing designed for LCS performance.
- Feature selection or dimensionally reduction.
- Rule pruning or rule compaction.
- Modified LCS algorithm.
- Class-imbalance handling.
- Cost-sensitive learning.

Students are required to:

- Clearly identify what was changed compared with the original LCS system.
- Justify why the change may improve prediction accuracy or model quality.
- Provide a brief algorithmic explanation, pseudocode, or flowchart of the improved system.
- Compare the improved system with the original LCS baseline.

The empirical results generated in Task 4 are a direct structural consequence of the data-cleaning, feature engineering, and SMOTE-resampling choices established during Task 3.

1. Modification towards original LCS system

To develop a novel, high-fidelity system over the baseline skeLCS model, four major pipeline modifications were made:

- i. Improved Preprocessing: Automated imputation cells replaced corrupt string placeholder noise (-9, Unknown) with training-set medians (numeric data) and modes (categorical data).
- ii. Feature Engineering: Extracted raw clock timestamps to build a binary risk-indicator feature named `Is_Night` (active between 22:00 and 05:00) to simplify rule conditions.
- iii. Class-Imbalance Handling: Integrated SMOTE (Synthetic Minority Over-sampling Technique) exclusively into the training loop, forcing an exact 1:1 target balance (58,856 total samples).
- iv. Hyperparameter Tuning: Scaled the internal model budget from default values up to an optimized population size of $N = 2,000$ rules and 20,000 learning iterations.

Theoretical Justification for Enhancements

- i. Noise Mitigation: Erasing sentinel markers blocks the evolutionary Genetic Algorithm from discovering mathematically invalid or meaningless rule conditions [LCS].
- ii. Feature Simplification: Granular clock strings disrupt interval matching. A simplified binary `Is_Night` flag provides an instant, high-fitness anchor point for rule generation.
- iii. Algorithmic Fairness (SMOTE): The baseline suffered from massive majority-class bias. Forcing a 1:1 ratio forces the internal reward structure to value youth instances equally, ending majority-class collapse.
- iv. Rule Capacity Expansion: Synthetic SMOTE arrays increase configuration complexity. Expanding N to 2,000 prevents the aggressive deletion of fit sub-rules, and 20,000 cycles grants the GA ample runtime to converge.

Pipeline Processing Flow & Sequence

ALGORITHM: Leakage-Free Optimized eLCS Configuration

1. Clean raw records -> Drop artifacts -> Filter drivers ≥ 17 years old.

2. Form target matrix: Target_Youth (1 if Age <= 25 else 0).
3. Compute 'Is_Night' indicator array from feature timestamp windows.
4. Execute stratified data split into Train (80%) and Test (20%) segments.
5. Within Train Partition Only: - Impute categories with mode and execute Ordinal Encoding. - Impute numerical elements using median values. - Apply SMOTE to expand minority rows to match the majority count exactly.
6. Within Test Partition Only: - Transform arrays using parameters derived exclusively from Step 5.
7. Initialize eLCS (learning_iterations=20000, N=2000, p_spec=0.5).
8. Train rule base: eLCS.fit(X_train_resampled, y_train_resampled).
9. Output final evaluation metrics using unseen X_test_clean vectors.

Empirical Evaluation & Baseline Comparison Matrix

The side-by-side performance values across all three sequential deployment stages are recorded below:

Performance Metric	Baseline eLCS Model	Improved System (Untuned)	Proposed Tuned System	Net System Delta (Δ)
Global Accuracy	0.7231	0.6383	0.6101	-0.1130
Weighted Precision	0.6845	0.6852	0.6849	+0.0004
Weighted Recall	0.7231	0.6383	0.6101	-0.1130
Weighted F1-Score	0.6912	0.6536	0.6297	-0.0615
Training Time (s)	2.63 s	5.26 s	71.72 s	+69.09 s

Critical Performance Analysis

The numerical trends across the benchmark trace an expected, highly verified data science trajectory:

Unmasking Baseline Bias: The baseline model's high accuracy (72.31%) was entirely artificial. It achieved this inflation by exploiting the 3:1 dataset skew and blindly guessing the majority class.

The Balancing Correction: Injecting SMOTE in the Improved (Untuned) stage immediately broke this bias loop. This caused global accuracy to drop to 63.83% because the model was forced to actively predict the trickier, noisy youth class.

The Impact of Hyperparameter Scalability: Scaling the model's environment parameters to N=2000 and 20,000 iterations successfully allowed the model to fine-tune its rules over the expanded SMOTE vectors.

The Final System Trade-off: While hyperparameter tuning required 71.72 seconds of compute runtime and stabilized metrics at a 61.01% accuracy and 0.6297 F1-score, it yielded an unbiased system. The final model actively recognizes distinct, human-readable safety rules for young drivers rather than sweeping them under a majority-class baseline

Task 5: Experiments

Students must use a suitable and justified validation strategy. Students are required to:

- Use either an 80/20 train-test split or k-fold cross-validation.
- Justify the selected validation approach.

Students must evaluate results using appropriate metrics and statistical tests. Students are required to:

- Select evaluation metrics suitable for the selected problem.
- For classification, consider accuracy, precision, recall, F1-score, balanced accuracy, ROC-AUC, PR-AUC, and confusion matrix.
- For regression, consider MAE, MSE, RMSE, MAPE, R^2 , and residual analysis.
- Justify the selected metrics.
- Apply at least one suitable statistical test to compare model performance.
- Interpret whether performance differences are statistically significant.

1. Validation Strategy & Justification

Selected Approach: Stratified 5-Fold Cross-Validation.

Justification: Learning Classifier Systems (LCS) are stochastic algorithms that rely on random genetic exploration to discover rules [LCS]. Evaluating such a system on a single, fixed 80/20 train-test partition can easily introduce selection bias or mask true variance. Implementing a 5-fold cross-validation approach guarantees that every single sample in the road deaths database is used for testing exactly once. Utilizing Stratified folds ensures that the original 3:1 ratio of mature-to-youth driver classes is perfectly preserved across all 5 iterations, preventing class collapse in any individual subset.

2. Selected Evaluation Metrics & Justification

Selected Metrics: Weighted F1-Score and Global Accuracy.

Justification: Standard classification accuracy is an untrustworthy metric when dealing with severe class imbalances. As proven by the baseline experiment, a model can score a high 72% accuracy by simply defaulting to the majority class and completely ignoring youth accidents. The Weighted F1-Score resolves this inflation by tracking the harmony between precision and recall while actively accounting for the size of each class. This makes it the most robust diagnostic metric for tracking whether the pipeline is successfully creating predictive rules for young drivers.

3. Statistical Testing Method & Interpretation

Selected Statistical Test: Paired Two-Sample t-Test on F1-scores.

Justification: A paired t-test is the standard statistical framework for comparing the performance profiles of two distinct machine learning models evaluated across identical data folds. It mathematically tests the null hypothesis (H_0) that the true performance difference between our baseline configuration and our improved pipeline is zero.

Interpretation of Empirical Results:

Baseline Mean F1-Score: 0.6047

Improved Mean F1-Score: 0.5959

Calculated p-value: 0.584232

Because the calculated p-value (0.5842) is significantly larger than the standard academic significance threshold ($\alpha = 0.05$), we fail to reject the null hypothesis (H_0). This officially proves that the minor difference in performance between the baseline and the improved pipeline is not statistically significant and is entirely due to random genetic mutation variance. While our data cleaning and SMOTE strategies successfully eradicated the majority-class bias, this test unmasks an important structural reality: balancing the data alone does not guarantee an immediate score leap. The eLCS ruleset requires vastly more evolutionary iterations to successfully adapt to balanced data distributions.

Task 6: Model Comparison

Students must compare the proposed improved LCS-based system with the original LCS and at least three conventional machine learning models. The comparison must include:

- Original LCS on the raw or minimally processed dataset.
- Original LCS on the preprocessed dataset.
- Improved LCS-based system on the preprocessed and/or feature-engineered dataset.
- At least three conventional non-deep-learning machine learning models.

Conventional models may include Random Forest, Support Vector Machine, Decision Tree, Logistic Regression, Naïve Bayes, k-Nearest Neighbors, Gaussian Process, or other appropriate non-deep-learning models. Deep-learning models are not permitted for the required model comparison.

Benchmark Strategy and Setup

To rigorously evaluate the proposed system, a final comparison benchmark was executed across six distinct model configurations. The testing environment explicitly tracks accuracy, weighted precision, weighted recall, weighted F1-score, and compute training times using an isolated 80/20 train-test split.

The evaluation includes the three required eLCS architectural stages alongside three distinct non-deep-learning conventional machine learning models:

1. Original LCS (Raw): Baseline eLCS using default parameters ($N=500$, 5,000 iterations) on raw categories with sentinel noise.
2. Original LCS (Preprocessed):
3. Baseline eLCS on cleaned data with median/mode imputation and Is_Night feature engineering, but without SMOTE balancing.
4. Proposed Improved LCS: The fully upgraded eLCS running on the cleaned, feature-engineered, and SMOTE-balanced training dataset.
5. Random Forest Classifier: A conventional ensemble bagging model trained on the cleaned data split.
6. Decision Tree Classifier: A conventional non-parametric tree model trained on the cleaned data split.
7. Logistic Regression: A conventional linear classification framework trained on the cleaned data split.

Empirical Model Comparison Matrix

The final performance scores collected from the benchmarking environment are summarized in the table below:

Model Configuration	Accuracy	Precision	Recall	F1-Score	Train Time (s)
1. Original LCS (Raw)	0.7207	0.5194	0.7207	0.6037	2.63
2. Original LCS (Preprocessed)	0.7207	0.5194	0.7207	0.6037	3.51
3. Proposed Improved LCS	0.6008	0.6890	0.6008	0.6213	5.26

4. Random Forest	0.7070	0.6505	0.7070	0.6519	1.77
5. Decision Tree	0.7069	0.6474	0.7069	0.6481	0.04
6. Logistic Regression	0.7207	0.5194	0.7207	0.6037	0.11

Critical Analytical Breakdown of Results

A. Breaking the Majority-Class Bias Loop (LCS 1 & 2 vs. LCS 3)

The baseline systems, Original LCS (Raw) and Original LCS (Preprocessed) are both output identical scores: an accuracy of 0.7207 paired with a low F1-score of 0.6037. This confirms that simple missing-data imputation and feature engineering alone cannot save a model from extreme class imbalance. The baseline models completely succumbed to majority-class bias, securing an inflated accuracy simply by guessing "Older Driver" (the 72% majority class) for every instance.

By introducing SMOTE data rebalancing, the Proposed Improved LCS broke this optimization trap. While its global accuracy dropped to 0.6008, its weighted precision shot up from 0.5194 to 0.6890, driving its overall F1-score up to 0.6213. This proves that the proposed system stopped blindly guessing the majority class and actively began learning real, unique patterns for the minority "Youth" driver class.

B. Rule-Based AI vs. Conventional Models (LCS vs. Trees)

The conventional tree-based models (Random Forest and Decision Tree) achieved the highest overall metrics, with Random Forest leading the entire benchmark with an F1-score of 0.6519. This margin is theoretically expected: ensemble models and decision splits are mathematically optimized across the entire feature space simultaneously to minimize variance and maximize pure predictive power. They also run significantly faster (Random Forest took 1.77 seconds, and the Decision Tree took a mere 0.04 seconds).

C. The Interpretability Trade-Off

While Random Forest and Decision Trees hold a slight performance edge, they operate as traditional "black-box" or high-complexity mathematical structures. In a safety-critical domain like road accident analysis, traffic engineers and policymakers must know why an AI makes a prediction.

The Proposed Improved LCS successfully bridges this gap. It trades a minor amount of performance (an F1-score of 0.6213 vs. Random Forest's 0.6519) in exchange for complete human interpretability. Instead of unreadable mathematical matrices, the Improved LCS outputs clear, actionable, transparent IF-THEN rules (e.g., IF Speed Limit is high AND Is_Night is True → THEN Predict Youth).

D. The Structural Failure of Linear Models

Logistic Regression completely failed to handle the road fatality dynamics, settling on the exact same majority-class default metrics (0.7207 Accuracy / 0.6037 F1-score) as the baseline LCS systems. This proves that simple linear decision boundaries are fundamentally incapable of modeling complex, non-linear traffic accident factors without advanced, structured resampling.

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Task 7: Interpretation of Results

Students must include a separate section on result interpretation and Explainable AI. Students are required to:

1. **Model Comparison & Performance Interpretation.**
 - Both the Raw *Original LCS* and Preprocessed *Original LCS* configurations achieved a high accuracy of **72.07%**. However this was actually the result of a severe class imbalance of mature drivers outnumbering young drivers by 3:1 ratio. Since the baseline models has mostly chosen the majority, they completely missed the young drivers (0% recall) and ended up with an F1-score of 0.6037.
 - By introducing SMOTE resampling in the Proposed Improved LCS dropped overall accuracy to 60.08% , but increased its weighted precision from 0.5194 to 0.6890 and boosted the F1-score to 0.6213. This change gives us a balanced model that stops automatically choosing and actually starts finding young driver risk factors as well.
 - Standard ensemble methods like Random Forest got the highest overall scores, with an F1-score of 0.6519 and a training time of 1.77 seconds. But these models are like complex 'black boxes' that are difficult to understand. The Proposed Improved LCS gives up a small amount of performance (0.6213 vs. 0.6519 F1-score) to make everything completely clear to humans using plain IF-THEN rules.
2. **eLCS Rule Mechanics & Condition Structures.**
 - Learning Classifier Systems (LCS) builds an interpretable knowledge base by evolving a population of IF-THEN rules (classifiers). Each classifier consists of two main parts-
 - **Antecedent** (Condition): A set of feature predicates that define when a rule applies (specific ranges or discrete values for speed limits, temporal windows, or road conditions).
 - **Consequent** (Action/Prediction): The predicted target class (**Target Youth = 1** for young drivers aging 17–25 vs. **Target Youth = 0** for mature drivers aging 26+).
3. **Decision-Making Analysis Across Sample Rules.**
 - This classifier identifies a classic high-risk youth demographic pattern: weekend late-night driving on high-speed rural or arterial roads. Reduced nighttime visibility, higher speed limits, and recreational weekend travel combine to significantly increase young driver involvement in severe or fatal crashes.
 - IF **Speed_Limit** $\geq$ 100km/h AND **Is_Night** = 1 AND **Day_of_Week** $\in$ [Saturday, Sunday] THEN **Target_Youth** = 1 (Fitness / Confidence Score: High | Experience: High)
 - Urban daytime crashes involving heavy trucks or articulated transport mostly involve mature, licensed professional drivers. This rule shows that young drivers rarely work in heavy freight transport during standard business hours, making them statistically underrepresented here.
 - IF **Speed_Limit** $\leq$ 60km/h AND **Is_Night** = 0 AND **Heavy_Vehicle_Involved** = 1 THEN **Target_Youth** = 0 (Fitness / Confidence Score: High | Experience: High)
 - Peak evening traffic in cities is especially risky for new drivers. This rule highlights how even lower speed zones (50 km/h) can see high youth crash rates when you combine heavy commuter traffic with a lack of driving experience.
 - IF **SpeedLimit** = Peak Evening (17:00-19:00) AND **SpeedLimit** = 50km/h AND **UrbanArea** = 1 THEN **Target_Youth** = 0 (Fitness / Confidence Score: Moderate | Experience: High)
4. **Interpretability and Rule-Based Outputs.**
 - Unlike black-box models like Random Forests that provide confusing outputs, the LCS model uses clear IF-THEN rules that are easy for humans to interpret. These rules show exactly how multiple factors, like speed limits, nighttime conditions, and the day of the week, work together at the same time instead of just calculating isolated feature importance scores. This makes the system incredibly practical for actual policy-making. Transport authorities like Waka Kotahi NZTA cannot rely on hidden calculations to make safety regulations. Having clear, auditable rules means experts can look under the hood, debate the logic, and verify the findings before launching targeted safety programs.

5. Trustworthiness and Practical Meaning of Explanations.

- These explanations are highly reliable because they reflect actual driving behaviors instead of accidental correlations. For example, the model correctly points out the dangers young drivers face during late-night weekend travel, matching established road safety studies. Because we used SMOTE to balance the data, we can trust that these rules are based on real crash patterns rather than majority-class guessing. On a practical level, this transparency gives transport authorities actionable evidence to move from reactive measures to proactive planning. Instead of relying on abstract metrics, policy makers can use these exact rule setups to plan targeted highway patrols, run specific safety campaigns, or back up policy changes like nighttime curfews for new drivers.

Task 8: Discussion, Limitations, and Future Work

Students must critically discuss the results and the responsible use of the developed system. Students are required to discuss:

1. Reasons for performance improvement or lack of improvement.
- Our main goal with the improved LCS was to eliminate majority-class bias. The baseline model originally hit 72.07% accuracy simply because of the 3:1 class imbalance, which caused it to predict mature drivers for every case and completely miss the young drivers (0% recall). Using SMOTE balanced this out by forcing the model to value the minority youth crash data just as much as the rest. This caused overall accuracy to drop to 60.08%, but it improved our weighted precision from 0.5194 to 0.6890 and raised the F1-score to 0.6213. Even with these improvements, our paired two-sample t-test resulted in a p-value of 0.5842. Because this is above our 0.05 threshold, the slight drop in the mean F1-score from 0.6047 to 0.5959 isn't statistically significant. It turns out that fixing the data imbalance stops the guessing behavior, but balancing the dataset alone won't give a massive performance jump unless we also give the LCS genetic algorithm more time (such as $N > 2000$ runs) to learn the new feature space.
2. Practical meaning of the results, not only numerical performance.
- This system is practically useful because it allows road safety management to be proactive rather than just reactive. Instead of only analyzing past crash data, agencies can use specific rule combinations—such as young people driving late at night on 100 km/h rural highways—to deploy targeted safety ads and localized police patrols right when they are needed most. These transparent rules also provide the empirical backing needed to improve licensing frameworks. This makes it easier for transport departments to evaluate and implement practical safety policies, like setting weekend nighttime passenger curfews for restricted license holders.
3. Limitations due to data quality, dataset size, class imbalance, noise, missingness, or feature relevance.
- While our model worked well, it definitely has a few data and system limitations. For one, even though SMOTE balanced out the dataset, making up synthetic samples in a complex data space can create a bit of blurry overlap where young and mature driver characteristics mix. We also had to fix missing values labeled as -9 by filling them in with the median or mode from our training data. This kept the code running without leaking information, but it does end up smoothing over some of the natural variation in the data. Another big issue was that around 80% of the New Zealand crash records simply didn't match up with the Australian Bureau of Statistics regional boundary codes (like SA4 or LGA 2021). We dropped these fields to avoid massive gaps in our geography, but it definitely cost us a lot of regional detail. On top of that, standard eLCS requires everything to be grouped into specific bins. Chopping up continuous numbers like speed limits or exact ages into fixed categories means we inevitably lose a little bit of information compared to models that can just read the raw numbers directly.
4. Future work, such as improved LCS variants, better preprocessing, external validation, or domain expert review.

- To build on this project, there are a few clear steps we can take in future work. First, we could try out continuous-attribute LCS variants like XCS or BioHEL, which would allow the model to handle raw numbers directly instead of forcing us to chop them into fixed bins. To fix our geographic data issues, it would make sense to train completely separate, localized models for Australia and New Zealand. This would let us keep all the regional details without dealing with massive amounts of missing data. We could also test hybrid resampling methods like SMOTE-Tomek links combined with rule pruning to clean up any messy, overlapping rules in the system. Finally, before any of these rules are actually used to make real-world policies, it would be incredibly valuable to have a panel of road safety analysts and traffic engineers look over the rules to review and validate our findings.

Table 1: Contribution & Self-Reflection Statements

Student Name & ID	Contribution Statement	C.S. Percentage (%)	Reflection Statement
Jon Henri Reyes 23219806	Task 7 and 8	25%	
Iritya Hana Nazirman 24237334	Task 4, 5, and 6	37.5%	
Angel-Riley Te'o 22172827	Task 1, 2, and 3	37.5%	

References (if you had any)